



D4 · DELIVERABLE

Audit Specification

Metrics, measurement frequency, thresholds and the breach response procedure for the recurring bias audit.

Turns "selection rate for any declared group must not fall below 80% of the highest group's rate" into a measurable, repeatable audit with a defined consequence, and reports the current results.

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AUTHOR
Anshuman Atrey
Roll 150096724029

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1 Purpose and scope

This specification defines the **recurring bias audit** of HireSense screening outcomes: what is measured, for which groups, how often, the thresholds, and the response procedure when a threshold is breached. It implements the case-study requirement:

"Selection rate for any declared group must not fall below 80% of the highest group's rate."

The audit covers automated and human screening decisions (shortlist / waitlist / reject) for every open role, and company-wide. It audits **outcomes**, not intentions. The screening routine never reads group attributes (D2 FR-S4), yet outcomes can still differ between groups through proxies. Catching that is the audit's purpose.

2 Declared groups and data sources

DIMENSION	GROUPS	SOURCE	HANDLING
Gender	Woman · Man · Non-binary · Prefer not to say	Voluntary self-identification at application	'Prefer not to say' is never compared
Age band	18–29 · 30–39 · 40+ · Prefer not to say	Voluntary self-identification	Same
University tier	Tier 1 · Tier 2 · Tier 3 · Unknown	Derived from the parsed university, for the audit only	Included because the internal review found the old ranking tracked tier; 'Unknown' not compared

Self-identification is stored in a separate `self_id` table. Only the audit module reads it. It is never shown to recruiters or returned by application APIs (D2 NFR-5).

3 Metrics

METRIC	DEFINITION	NOTES
Selection rate	$SR_g = \text{shortlisted}_g \div \text{decided}_g$	Decided = shortlist + waitlist + reject. Applications still in manual review are excluded until a person decides them.
Impact ratio	$IR_g = SR_g \div \max_{h \in C} SR_h$	C = comparable groups: $n \geq 30$ and not 'Prefer not to say'/'Unknown'. Small groups cannot set the benchmark.
Lowest impact ratio	$\min_{g \in C} IR_g$ per dimension	Reported as 'none' when fewer than two groups are comparable (insufficient sample).
Significance	$z = (SR_{\text{best}} - SR_g) \div \sqrt{(p(1-p)(1/n_g + 1/n_{\text{best}}))}$	Two-proportion z-test with pooled p ; $ z \geq 1.96 \Rightarrow$ the gap is unlikely to be sampling noise (5% level).
Sample size	n_g decided applications in the window	Groups with $n < 30$ are reported, not enforced.

4 Requirement and acceptance tests

Testable requirement (FR-F1)

For every declared group g in a dimension with $n_g \geq 30$ decided applications in the audit window: $IR_g \geq 0.80$. Any comparable group with $IR_g < 0.80$ is a **breach** of that dimension.

#	GIVEN	THEN	AUTOMATED TEST
AT-1	A: 100 decided, 50 selected · B: 100 decided, 40 selected	$IR_B = 0.80 \rightarrow$ pass (boundary is inclusive)	Pass
AT-2	A: 50/100 · B: 39/100	$IR_B = 0.78 \rightarrow$ breach	Pass
AT-3	A: 50/100 · B: 1/10	B is low-sample $\rightarrow$ reported, no breach	Pass
AT-4	A: 50/100 · 'Prefer not to say': 5/100	never compared	Pass
AT-5	A: 40 decided + 50 in manual review	rate uses the 40 decided only	Pass
AT-6	Only one group has $n \geq 30$	no ratio, 'insufficient sample'	Pass
AT-7	A: 20/40 · B: 15/40 (IR 0.75, not significant)	warning , job not paused	Pass
AT-8	A: 150/300 · B: 105/300 (IR 0.70, $z \geq 1.96$)	critical , auto-reject paused, the next rejection goes to review	Pass
AT-9	Paused job, re-audit passes	auto-reject resumes	Pass

AT-1 to AT-6 are in `audit.test.ts` and AT-7 to AT-9 in `audit-service.test.ts` (in-memory database, end to end).

5 Frequency, windows and triggers

TRIGGER	WHEN	SCOPE	WINDOW
Scheduled	1st of every month, 06:00	Each open role + company-wide	Rolling 90 days (enough sample per group)
Criteria change	Immediately after a new criteria version is published	That role	Last 90 days, re-screened under the new version
Pre-publication (preview)	Before publishing criteria	That role	All applicants, dry run (nothing recorded)
Parser / taxonomy release	After a new parser, OCR model or skill-taxonomy version	All roles	Last 90 days, re-parsed
On demand	Audit lead, complaint, or Head of Talent request	Any	Last 90 days
Escalated cadence	After a warning	That role	Weekly until two consecutive passes

Why a rolling 90-day window, not a calendar month? One role receives about 30 applications a month, which is too few for stable rates per group. In the demo data, monthly company-wide windows (~150 decisions) produced "critical" breaches from sampling noise alone. Ninety-day windows with $n \geq 30$ per group and a significance test separate real disparities from noise.

6 Thresholds and severity ladder

SEVERITY	CONDITION	AUTOMATIC ACTION	HUMAN ACTION AND DEADLINE
Pass	No comparable group below 0.80	Record audit; resume auto-reject if paused	—
Warning	Breach ($IR < 0.80$) but not significant ($z < 1.96$)	Open incident; raise this role's audit cadence to weekly	Criteria owner reviews the job-relatedness of the criteria within 5 working days
Critical	Breach and significant ($z \geq 1.96$)	Pause automated rejection for the role: every rejection is routed to human review	Audit lead and criteria owner start root-cause analysis within 2 working days ; Head of Talent informed

Company-wide breaches escalate to the Head of Talent and trigger per-role audits. They do not pause a role on their own.

7 Breach response procedure

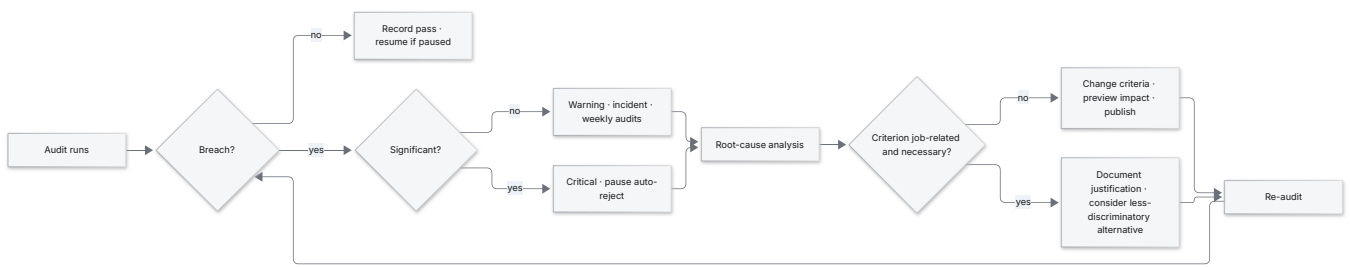


Figure 1. Breach response loop. Every step leaves a record in the audit table or the decision log.

STEP	WHAT HAPPENS	RESPONSIBLE	ACCOUNTABLE	RECORD
1 Detect	Scheduled or triggered audit computes ratios and grades severity	System	Audit lead	audits row (results, severity, action)
2 Contain	Critical: auto-reject paused for the role	System	Audit lead	jobs.auto_reject_paused + reason
3 Notify	Criteria owner and Head of Talent informed with the audit link	Audit lead	Head of Talent	Incident ticket
4 Analyse	Which criterion drives the gap? Compare by criterion; use the impact preview to test alternatives	Criteria owner	Audit lead	Analysis note
5 Remediate	Change criteria (new version, with a mandatory note), or document job-relatedness	Criteria owner	Head of Talent	criteria_versions row / justification
6 Verify	Re-audit (automatic on criteria change); two consecutive passes close it	System	Audit lead	audits rows
7 Review	Monthly governance review of open incidents and KPIs (D6)	Audit lead	Head of Talent	Minutes

8 Current results (seeded data)

AUDITS RECORDED 63 scheduled + triggered	LATEST COMPANY AUDIT critical 2026-06-01 → 2026-09-01	ROLES PAUSED 1 auto-reject paused now	REJECTIONS HELD FOR REVIEW 67 because of a pause
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Per-role status (latest scheduled audit)

ROLE	WINDOW	SEVERITY	LOWEST RATIO	AUTOMATED REJECTION
Senior Frontend Engineer	2026-06-01 → 2026-09-01	Critical	0.45	Paused
Backend Engineer (Java)	2026-06-01 → 2026-09-01	Warning	0.73	Active
Data Analyst	2026-06-01 → 2026-09-01	Pass	1.00	Active
DevOps Engineer	2026-06-01 → 2026-09-01	Pass	0.94	Active
QA Automation Engineer	2026-06-01 → 2026-09-01	Pass	—	Active
Sales Development Representative	2026-06-01 → 2026-09-01	Pass	—	Active

Finding 1: the legacy ranker's university-tier bias is gone

Company-wide, over all decided applications, on the same applicants in shadow mode:

GROUP	N DECIDED	SELECTED	IMPACT RATIO	Z	LEGACY RATIO
Tier 1	240	45%	1.00	—	1.00
Tier 2	593	41%	0.90	1.17	0.48
Tier 3	1071	41%	0.90	1.28	0.23
Unknown (n < 30)	2	0%	—	—	—

The legacy score added 30 points for Tier 1 and 12 for Tier 2. HireSense does not read university, and the Tier-3 impact ratio moves from **0.23** (legacy) to **0.90**.

Finding 2: experience bands act as an age proxy on a senior role

Senior Frontend Engineer, age band, all decided applications:

GROUP	N DECIDED	SELECTED	IMPACT RATIO	Z	LEGACY RATIO
18–29	259	36%	0.46*	7.69	0.28
30–39	113	80%	1.00	—	1.00
Prefer not to say (n < 30)	9	67%	—	—	—

The routine never sees age. Yet a threshold that needs 4+ years of experience selects the 18–29 band at a much lower rate, and the gap is significant (*). The audit marked the role **critical** and paused automated rejection. The next step under §7 is for the criteria owner to show the experience requirement is job-related, or to lower it. The impact preview (dry run, computed for this report) shows that raising the 2–4 year band's score to the threshold (75) moves the age-band ratio from 0.46 to **0.78**: better, but still below 0.80, so the change alone would not close the incident. A real disparity that the routine itself cannot see is exactly what a recurring audit exists to find.

* significant at the 5% level ($z \geq 1.96$). All candidates and outcomes are synthetic; the age–experience relationship is a property of how the demo data was generated.

9 Limitations

- **Self-ID non-response.** Groups exist only for applicants who choose to declare. Non-response is reported, and if it grows the audit loses power.
- **Multiple comparisons.** 3 dimensions × several groups × 7 scopes per month mean some "significant" breaches occur by chance. The seeded history shows two such false alarms (D6 §7). A Holm–Bonferroni correction before grading critical is the planned improvement.
- **Intersectional groups** (e.g. women aged 40+) are not audited in v1; they need larger samples. This is planned as a perfective maintenance item (D6).
- **Selection ≠ quality.** The 4/5ths rule measures adverse impact, not whether criteria predict job performance. Validity checks against later performance are a separate study.
- **Derived tier list** is illustrative and must be reviewed before production use.